Shifted face emoji in indirect discourse: A mixed-quotational approach[1]

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Abstract. *Face emoji* standardly receive an author-oriented interpretation (Grosz et al., 2021). Moreover, Grosz et al. (2023) claimed that they normally cannot receive a shifted interpretation in *indirect discourse* (ID). In this paper, we present the results of a rating study showing that emoji can receive a shifted interpretation, i.e., from the matrix subject's perspective, in ID utterances. Following a suggestion by Ebert and Hinterwimmer (2022), we advocate for an analysis of ID treating it as mixed quotation. Crucially, as we are dealing with multimodal data, we view quotation as an instance of demonstration (Clark and Gerrig, 1990).

Keywords: indirect discourse, mixed quotation, face emoji.

1. Introduction

Face emoji are a ubiquitous feature of digital communication, often expressing the author's affective stance toward their message. Previous research (e.g., Grosz et al., 2021, 2023) has claimed that face emoji are typically interpreted from the author's perspective and, crucially, do not undergo perspective shift in ID. However, our study challenges this assumption by demonstrating that face emoji can, under certain conditions, receive a shifted interpretation from the perspective of the matrix subject in ID utterances.

In this paper, we present the results of a rating study designed to test whether the interpretation of face emoji in ID can shift away from the speaker and toward the matrix subject when the latter's perspective is made prominent in the discourse. Our findings provide empirical support for such shifts, contradicting earlier claims that emoji are strictly author-bound. To account for these results, we propose an analysis of ID as a form of mixed quotation, following a suggestion by Ebert and Hinterwimmer (2022). Specifically, we argue that certain elements in ID, including face emoji, function as unmarked quotations that iconically depict aspects of the reported speech event. Our theoretical account builds on Clark and Gerrig's (1990) demonstration-based model of quotation and is grounded in a neo-Davidsonian event semantics. One of our guiding assumptions about speech events is that they have a form and a content component (Maier, 2017, see also Hacquard, 2010). We model quotation as similarity between the form of the reported speech event and the speech report, which we capture as a demonstration.

The paper is structured as follows: In Section 2, we provide background on speech reports, quotation, and the semantics of face emoji. Section 3 presents our experimental study, including methodology, results, and discussion. In Section 4, we develop our formal analysis, treating shifted emoji as demonstrations that iconically represent features of the reported speech event.

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Finally, Section 5 concludes with a discussion of broader implications and avenues for future research.

2. Background

2.1. Speech reports and quotation

In spoken and written communication, speakers have several options to indicate when they are reporting on someone else's thoughts or feelings:

(1) While waiting for the bus, Emma received a text from her friend inviting her to a concert.
  a. She replied, "I will buy a ticket later today."
  b. She replied that she would buy a ticket later that day.
  c. She would buy a ticket later today.

The sentence in (1a) is an instance of what is commonly referred to as *direct discourse* (DD). Typically marked by quotation marks in written communication, it renders an utterance of another individual from their perspective. In other words, the complete sentence enclosed by quotation marks is a verbatim repetition of what the reported speaker (= Emma) said or thought, and thus their perspective is conveyed exclusively. Consequently, all indexical expressions—in the case at hand the first-person pronoun *I*, the tense marking, and the temporal adverbial *today*—are not interpreted in the current utterance context, but receive a so-called shifted interpretation: they are interpreted with respect to the context where Mary replied to her friend and not the current utterance context. This means that, for example, *I* does not refer to the current speaker, but to Emma instead.

The type of speech report in (1b) is an instance of so-called *indirect discourse* (ID). Here, in contrast to DD, the reported speaker's as well as the current speaker's perspectives are conveyed. This is evidenced by the observation that indexicals and, more generally, perspective-dependent expressions, do not behave in a uniform manner in ID. One can notice, for example, that in ID a third-person pronoun is needed to refer to the reported speaker, thus contrasting with DD utterances. This pattern is attested not only in English, but also in German, Spanish, and many other languages (but see, for example, Schlenker, 2003; Anand and Nevins, 2004 for languages that allow for *indexical shift*). Moreover, tense markings and temporal adverbials are normally interpreted from the speaker's perspective. Thus, in the ID utterance, the indexical *today* has been replaced with the non-indexical *that day*. Note, however, that this is only a preference. In particular, temporal and local indexicals can receive a shifted interpretation if the perspective of the reported speaker is prominent in the surrounding discourse (Plank, 1986; Anderson, 2019). Due to the often times higher prominence of the speaker's perspective in ID (among other things), it is normally not analyzed as quotational. This, as we will show below, however, is difficult to reconcile with the results from the study reported in Section 3 and when taking into account multimodal data in general (e.g., Walter, 2025a). As we will argue, an analysis of ID should at least allow for certain elements to be quoted.

According to the standard analysis proposed by Hintikka (1969), propositional attitude verbs like *say* or *believe* are universal quantifiers over possible worlds that are compatible with what the respective attitude holder says or believes in the world of evaluation such that in all those worlds the proposition denoted by the complement clause is true.

Finally, the speech report type in (1c) is an instance of *free indirect discourse* (FID)—a means to report what someone else said or thought without any overt marking—, which is largely constrained to narrative discourse. It can be seen as a blend of ID and DD, since it exhibits properties of both speech report types because all perspective-dependent expressions except pronouns and tense markings are interpreted from the perspective of the reported speaker. Therefore, some analyses treat as similar to ID (e.g., Sharvit, 2008), whereas others treat it as more similar to DD (Schlenker, 2004; Maier, 2015, 2017). The latter view has been most radically defended by Maier (2015), who argues for an analysis of FID as a special, but conventionalized instance of mixed quotation where the whole utterance is quoted from the reported speaker with the systematic exceptions of pronouns and tense markings. This unquotation mechanism is driven by pragmatics.

Before turning to arguments in favor of treating quotation as an instance of demonstration (Clark and Gerrig, 1990), we would like to sketch briefly what *mixed quotation* (sometimes also called hybrid quotation) is. Mixed quotation is often defined as using quotation marks to simultaneously use and mention a phrase (Davidson, 1979). This is grounded in the assumption that mixed quotations make two independent meaning contributions (e.g., Potts, 2007): the first one can be retrieved when ignoring the quotation marks and is called the *use component*. The second meaning contribution of a mixed quotation, the *mention component*, is that the phrase inside the quotation marks is a verbatim repetition of the reported speaker's utterance. The example in (2) illustrates this.

(2) Alice said that life "is difficult to understand".

(Cappelen and Lepore, 1997: 429)

Here, the use component is that Alice uttered the proposition expressed by the clause under *said* in (2). The mention component is that she used the words *is difficult to understand* whilst expressing that proposition.

There are several families of semantic theories of quotation, among them the so-called name-based theory as originally proposed by Quine (1940) and the demonstrative theory as proposed by Davidson (1979). To illustrate the core assumptions of each theory, consider the example below.

(3) This morning John said "I am happy."

(Davidson, 2015: 482)

As the name implies, name-based theories treat quotations as names. A paraphrase of (3) would thus be something along the lines of *John said a sentence, the name of which is "I am happy"*. What is crucial to name-based theories is the assumption that the quotation marks are assigned the semantic function of turning whatever stands between the quotation marks into a name. While they share with name-based theories of quotation that they assign a crucial semantic role to quotation marks, demonstrative theories of quotation assign them the role of a covert demonstrative pointing to its referent, i.e., the quoted utterance. Thus, these two lines of analysis, and, more generally, all purely semantic theories of quotation assign a key semantic function to quotation marks. They have therefore been argued to make the so-called *necessity claim* (De Brabanter, 2023), meaning that all semantic theories assume that no quotation is possible without quotation marks.

While ascribing a crucial role to quotation marks might seem appealing and neat for semantic theories of quotation—especially when only considering written data—the reality is, unfortunately, not that simple. First, quotation is not a purely written phenomenon, but occurs in spoken (and signed) languages as well. Importantly, there is no spoken language equivalent that marks quotation as unambiguously as quotation marks do in written language (cf. De Brabanter, 2017; Johnson, 2017 and references cited therein). In addition, instances of (mixed) quotation have been identified that are not marked by quotation marks (Kirk-Giannini, 2024). For example, metalinguistic negation has been analyzed as an instance of unmarked mixed quotation (Potts, 2007). FID and protagonist projection have also been argued to be instances of unmarked mixed quotation (Maier, 2015; Stokke, 2021). A clear instance of unmarked mixed quotation occurring in a literary text is given in (4).

(4) To which Mr Bailey modestly replied that he hoped he knowed wot o'clock it wos in gineral.
(Charles Dickens, *Martin Chuzzlewit*, cited from Clark and Gerrig, 1990: 791)

Not only does the ID utterance foreshadow what we are going to argue for in this article, namely that ID allows for mixed quotation, but it also illustrates the use of a non-standard dialect that is not attributed to the speaker, but rather to the subject of the propositional attitude verb, i.e., Mr Bailey. In other words, the non-standard dialect is quoted from Mr Bailey. This phenomenon is known as (sentence-internal) language shift (Recanati, 2001). Kirk-Giannini (2024) argues that, consequently, (4) and (5) are semantically equivalent.

(5) To which Mr Bailey modestly replied that he hoped he "knowed wot o'clock it wos in gineral".
(Kirk-Giannini, 2024: 6)

We take the semantic equivalence of the two examples above as a striking argument against theories assuming the necessity claim. We will adopt a more pragmatic view of quotation in general and quotation marks in particular. Therefore, we assume that the presence of quotation marks is not obligatory for quotation. However, we assume that quotation must be (pragmatically) marked in some way, and that quotation marks are the most conventionalized way to do this in written language.

Moreover, the often assumed verbatimness requirement of quotations, i.e., that they are exact repetitions of what someone else said or thought, again only seems to hold in written language quotations and, as we will show in more detail below, not necessarily in all instances of written language quotation.

(6) I- I've only been- we've only been to like . four of his 1- five of his lectures, right?
(Clark and Gerrig, 1990: 795)

This utterance was quoted as stated in (7).

(7) Sidney said 'well I've only been to like four or five of his lectures'
(Clark and Gerrig, 1990: 796)

Theories positing a (strong) requirement of verbatim repetition for quotation cannot account for data as in (7). Additionally, the results of the experiments reported in Wade and Clark (1993) suggest that in spoken language quotations speakers are fully aware that their quotes are neither verbatim nor do addressees expect them to be verbatim reproductions of the original speech event.

The arguments above lead us to adopt a more pragmatic view of quotation. Specifically, we assume that quotation involves demonstration, which is, in other words, an iconic depiction of (parts of) a speech event. These demonstrations depict their referent from a certain viewpoint (Clark and Gerrig, 1990: 767). Clark and Gerrig (1990) argue that demonstrations consist of four aspects: i) *depictive* aspects, ii) *supportive* aspects, iii) *annotative* aspects, and iv) *incidental* aspects.

Depictive aspects are those parts of a demonstration which the speaker intends to resemble their referent (= the quoted utterance). This can be words, gestures, intonation, and facial expressions, but also non-linguistic parts of an event.

Supportive aspects are those parts of a demonstration which are not intended by the speaker to depict, but are nevertheless needed to perform the depictive aspects. For example, when depicting someone playing tennis, a speaker can make use of slow motion movements in order to depict the way in which someone else played tennis.

Turning to annotative aspects of a demonstration, these are “added as commentary on what is being demonstrated” (Clark and Gerrig, 1990: 768). This can involve, for example, that the demonstrating speaker smiles during the demonstration, but smiling was not part of the event that is demonstrated. The speaker performing the demonstration expects the addressee to recognize that these parts are not part of the depictive component.

Finally, incidental aspects of a demonstration are whatever remains of the demonstration, i.e., aspects a demonstrating speaker “has no specific intentions about” (Clark and Gerrig, 1990: 768). This can involve, for example, the speaker staggering a little because they are drunk (De Brabanter, 2017).

Although we acknowledge that it often is not very clear which parts of a demonstration belong to which aspect of the demonstrational act, we still posit that this approach to quotation is most suitable to capture not only shifted face emoji in ID, but also to account for shifted interpretations of temporal and local deictic adverbials and other material that can optionally receive a shifted interpretation in ID. Still, we want to highlight at this point that future research should investigate which parts of a demonstration are normally classified and, even more importantly, recognized by the addressee as being either depictive, supportive, annotative, or incidental aspects, as this would broaden the understanding of quotation in general and consequently allow for descriptively more adequate formal theories of demonstration and quotation.

In formal semantic theory, the view that quotation is a depictive act has been most explicitly defended by Davidson (2015). She draws evidence from quotative *be like*-constructions in spoken language and role shift phenomena in sign language. As this paper is concerned with spoken language ID utterances, we will focus on the evidence she provides from spoken language in the following. Davidson (2015) observes that *be like*-constructions are often highly depictive. While *be like* can demonstrate speech events, as in (8), it can also demonstrate non-linguistic actions (cf. (9)).[2]

(8) John was like "I'm soooo hungry."

(9) My cat was like "feed me!"

(Davidson, 2015: 485)

The sentence in (8) depicts an event of John saying *I'm so hungry* where *so* is lengthened, which is indicated by the letter replications in the example. This means that here, not only John's words are part of the quote, but also parts of his intonation are quoted. The sentence in (9) is intended to depict a situation where the speaker's cat behaves a certain way, suggesting that it is hungry. This, however, is a clear instance of non-linguistic behavior, as cats are unable to produce the words embedded under *be like*. Still, the example is perfectly acceptable, even though it is not an utterance that is being quoted by the speaker. A strict semantic treatment of (9) assuming a verbatimness requirement could not account for this.

To formally model this, Davidson (2015: 487) proposes to add the demonstration type *d* to the semantic ontology. Adopting a neo-Davidsonian event semantics, she proposes a lexical entry for quotative *like* along the lines of (10).

(10) $[\![\text{like}]\!] = \lambda d \lambda e.[\text{demonstration}(d, e)]$

In formal terms, a demonstration is defined as follows (Davidson, 2015: 487):

(11) A demonstration d is a demonstration of e (i.e. demonstration(d, e) holds) if d reproduces properties of e and those properties are relevant in the context of speech.

Thus, in other words, a demonstration *d* of an event *e* is true, iff *d* is similar in certain features to *e*. The properties which can be deemed relevant for speech events include, among other things, words, intonation, facial expressions, or gesture. While we largely agree with this assumption, we will formally spell this out differently, and, as we believe, more adequately, in Section 4 below.

The lexical entry in (10) first combines with a demonstration, i.e., whatever is embedded under *like*. This is simply abbreviated as d_1 in the formalism and involves the words used for the demonstration and potentially other things, such as intonation in the case of (8). Next, *like* + demonstration combines with the verb *be*, thereby introducing an individual argument specifying that the individual saturating this argument (John in (8)) is the agent of the reported event. Finally, after λ-conversion and existential closure (Heim, 1982), this leaves us with the following semantics for (8) (Davidson, 2015: 487).

(12) $\exists e[\text{agent}(e, \text{john}) \wedge \text{demonstration}(d_1, e)]$

Thus, (8) is true, iff there exists an event with John being the agent and d_1 demonstrates this event in contextually determined aspects.

[2]Quotation marks in (8) and (9) are only inserted for purposes of illustration in these examples. Note that they would be equally acceptable if the quotation marks were omitted.

2.2. Some notes on (face) emoji

Face emoji are ubiquitous in digital communication and are therefore an important part of multimodal discourse, expressing affective attitudes towards the text they accompany (e.g., Kaiser and Grosz, 2021; Grosz, to appear). Intuitively, they resemble facial expressions revealing the emotions that speakers experience while making their utterances. This analogy is taken quite literally by (Maier, 2023), who assumes face emoji to be highly schematic little pictures of the facial expression the author has while typing the message accompanied by the respective emoji. The semantic contributions of pictures are analyzed along the lines of Greenberg (2013, 2021), i.e., as projections from parts of possible worlds as seen from a certain viewpoint to some surface. For face emoji, the canonical viewpoint is one facing the author of the message, and the projection function is highly stylized and schematic, ignoring most features of the actual facial expression being projected. A different analysis is proposed by Grosz et al. (2023), who assume face emoji to have a fixed, conventionalized denotation similar to expressives such as *damn* (which, originally may still be grounded in a resemblance relation between the respective emoji and the facial expression humans typically have when experiencing a particular emotion): They can only be used felicitously in a context *c* if the author of *c* has a certain emotional stance towards the proposition denoted by the message that the face emoji accompanies, where it is fixed for each emoji what that emotional stance is. The formal analysis to be proposed in Section 4 below is more easily compatible with the analysis proposed by Maier (2023), but in principle it could also incorporate the analysis proposed by Grosz et al. (2023).

In a similar vein as other iconic enrichments, such as (iconic) gestures (cf. Ebert and Ebert, 2014; Ebert et al., 2020; Schlenker, 2018) and ideophones (e.g., Henderson, 2016; Barnes et al., 2022; Barnes and Ebert, 2023), face emoji have been argued to contribute not-at-issue meaning by default (Grosz et al., 2023). They are thus not the main part of an utterance but can be seen as an aside the speaker does not intend to steer the conversation toward (Potts, 2005). A key property of not-at-issue content is that it cannot be directly denied in discourse. Instead, a discourse interrupting element, such as *Hey, wait a minute!*, has to be used (Shanon, 1976; von Fintel, 2004). Alternatively, one can challenge not-at-issue content by assenting with the proposition expressed by the at-issue content and then challenging the not-at-issue content (cf. Tonhauser's, 2012 diagnostic for not-at-issue content #1c). The examples in Figure 1 illustrate the non-deniability of emoji.[3]

However, if interlocutor B were to target the affective content of the emoji by means of something along the lines of *Yes, (I know), but you were grumpy AF*, this would make their response to interlocutor A more natural.

Moreover, it has been noted that face emoji are perspective-dependent in their interpretation and that they are by default interpreted from the perspective of the author (Kaiser and Grosz, 2021, Grosz et al., 2023, Grosz, to appear). This is also illustrated by the example in Figure 1. Here, the most natural interpretation of the emoji used by interlocutor A is from their own perspective. In other words, the use of the emoji comments on the proposition that A has just woken up that A is in a positive emotional state because of this. Experimental research, however, has shown that emoji can receive a shifted interpretation, i.e., a non-author-oriented

[3]Due to typesetting reasons, examples containing emoji will be shown in figures throughout the article. If displaying the emoji in a figure is not possible, it will be paraphrased.

Figure 1: Emoji cannot be directly denied in discourse. Examples taken from (Grosz et al., 2023: 309).

interpretation, when they co-occur with psych verbs (Kaiser and Grosz, 2021). The examples in Figure 2 illustrate this.

Figure 2: Emoji can receive shifted interpretations. Examples taken from Grosz (to appear: 14).

In the first example, the annoyed face emoji co-occurs with a sentence containing the object-experiencer verb *annoy*. The second example contains a subject-experiencer verb *admire*. In both cases, Kaiser and Grosz (2021) found that the holder of the affective attitude expressed by the emoji can shift from the author toward the experiencer (Adrian and Daniel, respectively). However, although this is dispreferred, their data show that the emoji can still be interpreted from the author's perspective.

It has been explicitly argued that face emoji cannot receive a shifted interpretation in ID, that is, an interpretation from the perspective of the matrix subject (Grosz et al., 2023: 909). However, consider the example in Figure 3. Here, intuitively, the pouting face emoji—under its most salient interpretation—expresses the attitude of Peter and not the attitude of the author of the message. Thus, against the claims made by Grosz et al. (2023), we hypothesize that face emoji can receive a shifted interpretation in ID. Acknowledging that they are normally strongly anchored to the perspective of the author and based on the findings on the shiftability of temporal and local deictic expressions discussed in Section 2.1 of the present paper, we furthermore hypothesize that face emoji can shift in ID only if the perspective of the matrix subject is prominent on the speech level. To test this, we conducted the rating study reported in the next section.

Figure 3: A shifted interpretation of an emoji in ID.

3. Experimental study

3.1. Participants

Thirty native German speakers participated in the study. They were recruited through Prolific[4].

3.2. Materials and procedure

We adapted the materials used for the study reported in Walter (2025a). Each utterance consisted of a sentence rendered in ID that was accompanied by a co-text face emoji, i.e., an emoji occurring at the end of the message (for a full typology of emoji-text alignment, see Pierini, 2021). We manipulated whether the speaker's or the matrix subject's (= Pia in (13)) perspective was prominent on the speech level (factor PERSPECTIVE: speaker vs. matrix subject). This was tested in a pilot study. Crucially, the face emoji could only receive a plausible interpretation from the matrix subject's perspective. An example is given in (13).

(13) a. **Matrix subject:** Pia fragte sich, warum ihre beste Freundin Anna, *diese gottverdammte Saufziege*, gestern Abend mal wieder ihr zu viel Wein nachgeschüttet hat, obwohl sie doch so wenig verträgt. + nauseated face
'Pia wondered why her best friend Anna, *that damned lush*, had poured her too much wine again last night, even though she couldn't handle it. + nauseated face'

b. **Speaker:** Pia fragte sich, warum ihre beste Freundin Anna, *die aber eigentlich nur die letzte Pfütze aus der Weinflasche loswerden wollte*, gestern Abend mal wieder ihr zu viel Wein nachgeschüttet hat, obwohl sie doch so wenig verträgt. + nauseated face
'Pia wondered why her best friend Anna, *who was just trying to get rid of the last drops in the bottle*, had poured her too much wine again last night, even though she couldn't handle it. + nauseated face'

We highlighted the perspective of the matrix subject or speaker by the appositives printed in italics in (13). When comparing them, one will notice that in (13a) the appositive's content most likely reflects Pia's attitude toward Anna, and not the speaker's. In contrast, the appositive in (13b) conveys information that is most plausibly interpreted from the current speaker's point of

[4]https://www.prolific.com

Speaker's perspective prominent　　Reported speaker's perspective prominent

Figure 4: Sample item in both conditions (translated).

view because Pia would most likely not be angry at Anna if she knew that she did not intend to make her drunk, but only to finish the wine bottle. Moreover, note that the nauseated face emoji (cf. also Figure 4) is also most plausibly interpreted from Pia's perspective.

To test for our hypothesis that face emoji can receive a shifted interpretation in ID if the matrix subject's perspective is prominent on the speech level, we constructed 24 experimental items that were distributed evenly onto four lists according to a Latin Square design and interspersed with 24 filler items. To make the experimental setting more natural, the items were displayed as text messages, as shown in Figure 4. Participants had to rate on a 7-point Likert scale the acceptability of these text messages. In order to familiarize participants with the experimental set-up, there was a training session consisting of two items after they had read an introductory text and gave informed consent. Based on our hypothesis, we predicted a main effect of PERSPECTIVE.

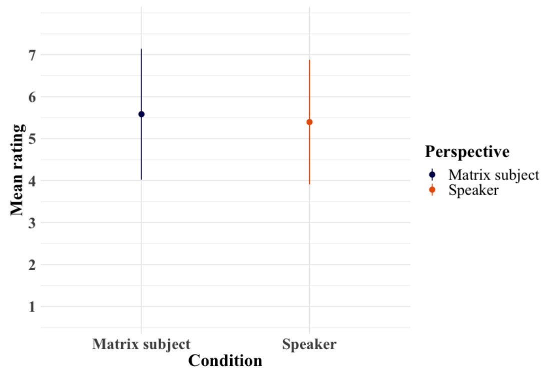

Figure 5: Mean values and standard deviations for each condition.

3.3. Results and Discussion

Data were analyzed using R statistics software (R Core Team, 2022). For data preparation and visualization, the tidyverse package (Wickham et al., 2019) was used. We tested for significant effects by analyzing the results using a cumulative link mixed effects model with the clmm() function of the ordinal package (Christensen, 2023). Stimuli, raw data, and analysis scripts can be found here: https://osf.io/a82cg/.

The mean ratings and standard deviations are shown in Figure 5. Items were overall rated slightly better when the matrix subject's perspective was prominent (M = 5.58, SD = 1.56) compared to when the speaker's perspective was prominent (M = 5.4, SD = 1.49).

Table 1: Ordinal mixed effects model with PERSPECTIVE as fixed effect and participants and items as random intercepts. Significance codes: *** 0.001 | ** 0.01 | * 0.05 | . 0.1

	Estimate	Std. Error	z value	Pr(> |z|)
Formula: *rating ~ perspective + (1 | participant) + (1 | item)*				
Perspective	0.348	0.175	1.99	0.047 *

The output of the ordinal mixed effects model is given in Table 3.3. It shows a main effect ($p < 0.05$) for PERSPECTIVE, suggesting that the tendency observed in the descriptive statistics is borne out: items were indeed rated better when the perspective of the matrix subject was prominent on the speech level compared to the perspective of the speaker.

The main effect of PERSPECTIVE confirms our hypothesis that face emoji can receive a shifted interpretation in ID, especially when the perspective of the matrix subject is prominent on the speech level. As will be elaborated on in more detail in the next section, we analyze shifted interpretations of face emoji in ID as instances of mixed quotation.

4. Formal proposal

As mentioned in the last section already, we informally propose to analyze shifted face emoji as instances of mixed quotation, or, more specifically, as a quoted facial expression from the reported speaker. Crucially, shifted face emoji are cases of unmarked quotation (e.g., De Brabanter, 2023; Kirk-Giannini, 2024), as there are no quotation marks present to mark the emoji that is being quoted.

We argue that, roughly speaking, an ID utterance containing quoted elements contributes meaning in multiple dimensions (Potts, 2007): i) a descriptive contribution in the at-issue dimension that there exists a reported speech event and ii) a depictive, i.e., iconic, contribution in the not-at-issue dimension stating that the quoted element is similar to parts of the reported speech event in contextually determined aspects. We follow Clark and Gerrig (1990) in assuming that quotation is depictive. We will model this by implementing the idea laid out in Davidson (2015) that demonstrations are part of the semantic ontology. Consequently, shifted face emoji are captured as demonstrations in our account.

To illustrate our proposal that mixed quotation in ID makes a multidimensional meaning contribution, we will first give some background on not-at-issue content. While at-issue content can be seen as the main point of an utterance (Potts, 2005), not-at-issue content is often seen as an aside or additional information the speaker does not wish to discuss. These differences result in different semantic behavior of not-at-issue content compared to at-issue content. For example, not-at-issue content projects through negation and other semantic operators (e.g., Langendoen and Savin, 1971; Heim, 1983; Geurts, 1999; Potts, 2005). In addition, not-at-issue content does not carry information relevant to the current question under discussion (Simons et al., 2010). Moreover, as touched upon already in Section 2.2, it cannot be directly denied in discourse.

If an interlocutor wants to target not-at-issue content, they have to make use of a discourse-interrupting protest like *Hey, wait a minute!*, for example (Shanon, 1976; von Fintel, 2004). Alternatively, not-at-issue content can be targeted by an adversative continuation following an assent, as in (14) (cf. Tonhauser's, 2012 diagnostic #1c). Crucially, this exclusively works to challenge not-at-issue content. When employing the same strategy to challenge at-issue content, this results in infelicity.

(14) A: My friend Sophie, a classical violinist, performed a piece by Mozart. (Syrett and Koev, 2015)
B_1: Yes, true, but she's not a classical violinist.
B_2: #Yes, true, but she did not perform a piece by Mozart.

The responses by B_1 and B_2, respectively, illustrate this. The response by B_1 is felicitous because it targets the proposition that Sophie is a classical violinist expressed by the appositive, which by default contributes not-at-issue meaning (Potts, 2005). Conversely, as evidenced by the response of B_2, the adversative continuation cannot target the at-issue proposition expressed by the main clause in A's utterance.

Transferring this to the at-issue/not-at-issue divide proposed above for ID, consider the following utterances:

(15) A: Peter complained that he again had to pay the bill for the whole group. + pouting emoji
B_1: Yes, true, but he didn't look this angry when saying it.
$B_{2/3}$: #Yes, true, but he didn't complain/it wasn't Peter who said that.

Although admittedly, the utterance of B_1 is highly marked as responses to A in (15), one can imagine contexts (for example, in some kind of rehearsal context) where this dialogue would work. What is central to our argument here is that B_1's utterance targets part of the depictive component of the ID utterance, i.e., the quoted parts of the utterance. Since it can be targeted by an adversative continuation following an assent, we take this as evidence in favor of our assumption that the quotational component in an ID utterance makes a default not-at-issue contribution. Turning next to the responses of B_2 and B_3, it is important to note that they intend to target either i) the type of speech event that the ID utterance exemplifies—i.e., that it was a complaining event—or ii) who the reported speaker was. In other words, both responses aim to target what we hypothesize to be part of the descriptive at-issue, or, in other words, non-quotational component of the ID utterance. Since both continuations are infelicitous responses to A in (15), we take this as further evidence in favor of our proposed at-issue/not-at-issue divide in ID. To represent this in our formalism, we make use of the propositional variables p and $p*$ to represent at-issue and not-at-issue content, respectively (cf. Ebert et al., 2020).

For our analysis, we follow the proposal made in Kratzer (2006, 2016) that propositional attitude verbs are essentially transitive verbs taking a special type of direct objects as its argument, namely belief objects (see also Maier, 2020 for an extension to quotation). She embeds this proposal in a neo-Davidsonian event semantics, thereby rejecting the traditional Hintikka-style treatment (Hintikka, 1969) of verbs of saying as intensional operators. We also assume that speech events, in contrast to other event types, have a form and a content component beside the usual event properties such as agent, theme, etc. (Maier, 2017), where the content is the proposition expressed by the respective speech event. Maier (2018: 269) makes use of this distinction in his analysis of quotation, where quotation marks have the denotation in (16).

(16) "…" $\rightsquigarrow$ $\lambda q \lambda e.[\text{form}(e, q)]$

Thus, quotation marks denote a function taking an event and an utterance as its arguments and imposing the form restriction that the quoted constituent has the same form as the reported speech event. Consider (17) for illustration (Quine corners are used to indicate that reference to something of the utterance type *u* is made, cf. Potts, 2007):

(17) a. Daniel said "I'm hungry".
b. $\exists e[\text{say}(e) \wedge \text{agent}(e, \text{daniel}) \wedge \text{form}(e, \ulcorner\text{I'm hungry}\urcorner)]$

Thus, according to (17b), the DD utterance in (17a) is true, iff there exists an event such that it was a saying event, with Daniel being the agent of this event and the form of the event was ⌜I'm hungry⌝, thereby enforcing the often assumed verbatimness requirement for DD.

Conversely, ID utterances do not make reference to the form of the saying event, but to the content of the saying event. This can roughly be formalized as in (18) (cf. Maier, 2017). The lexical entry of the propositional attitude verb *say* is thus as in (18c).

(18) a. Daniel said that he is hungry.
b. $\exists e[\text{say}(e) \wedge \text{agent}(e, \text{daniel}) \wedge \text{content}(e, [\![\text{he is hungry}]\!])]$
c. $[\![\text{say}]\!] = \lambda p \lambda x \lambda e.[\text{say}(e) \wedge \text{agent}(e, x) \wedge \text{content}(e, p)]$

Thus, the main difference between DD and ID in the account by Maier (2017, 2018) pertains to the component of the saying event each speech report type makes reference to: while DD imposes a form restriction by making reference to the form component of the utterance event, ID makes reference to the propositional content of the saying event. In other words, ID imposes a content restriction on the speech event it describes.

A shortcoming of lexical entries for verbs of saying along the lines of (18c) is that it does not straightforwardly capture shifted perspective-dependent expressions in ID, such as shifted face emoji (shown above), shifted temporal and local adverbials (Plank, 1986; Anderson, 2019), or self-pointing gestures aligned with a third-person pronoun that receive an interpretation from the perspective of the matrix subject (Ebert and Hinterwimmer, 2022; Walter, 2025a, b). To capture these phenomena, normally some context shift or shifting operator (Anand and Nevins, 2004) would be assumed. We want to propose, however, that these phenomena can be captured in a non-monstrous way—i.e., by analyzing such cases as instances of unmarked mixed quotation (Kirk-Giannini, 2024). Crucially, in order to enable our proposal to make predictions about spoken quotation, which is often times a multimodal phenomenon (e.g., Stec, 2012, 2016), as well, we view quotation as an iconic act, namely a demonstration (Clark and Gerrig, 1990). To spell this idea out formally, we incorporate Davidson's (2015) idea of treating demonstrations as part of the semantic ontology denoting objects of type *d* (cf. Section 2.1 of the present paper).

We argued above that the ID demonstration of the reported speech event and the form of a subevent of the reported speech event need to be similar from a certain viewpoint (see also Schlenker and Lamberton, 2024 for a similar analysis of classifiers in sign languages). We propose to model this by adapting the not-at-issue similarity predicate SIM_{p*} proposed by Ebert et al. (2020) to compare the form of an event with the demonstration contained in the ID utterance. This makes an interesting prediction concerning the ontological status of demonstrations. As argued by Umbach and Gust (2014), similarity can only be established between objects of the same semantic type. Consequently, our proposal implicitly encodes the idea that demonstrations are a special type of events. While intuitively this seems appealing and on the right track (cf. Davidson, 2015 for similar remarks), as to the best of our knowledge only instances of event demonstrations have been discussed in the literature, so far (Clark and Gerrig, 1990; Wade and Clark, 1993; De Brabanter, 2017; Steinbach, 2023, among others). We cannot discuss this in detail, as it exceeds the scope of the article. However, we would like to emphasize here that future research should investigate the semantic and pragmatic status of demonstrations.

Before turning to the formalism, we would like to briefly emphasize that we take gestures, special types of intonation, as well as facial expressions to be subevents of a speech event. If one of these elements receives a shifted interpretation, or, in our terms, an interpretation as a quoted element (= a demonstration), we argue that this demonstration must be similar to a subevent of the reported speech event (e'' in what follows). The content restriction has to hold between the embedded proposition (p') and the remaining part of the speech event, e'. Consider now the sentence in (19) (repeated from Figure 3).

(19) Peter complained that he again had to pay the bill for the whole group. + pouting emoji

For the propositional attitude verb *complain*, we propose the following lexical entry:

(20) $[\![\text{complain}]\!] =$
$\lambda p' \lambda d \lambda x \lambda e.[\text{complain}_p(e) \wedge \text{agent}_p(e,\ x) \wedge\ \exists e' \subseteq e\ \exists e'' \subseteq e[\text{content}_p(e',\ p') \wedge$
$\text{SIM}_{p*}(\text{form}(e''),\ d)]]$

After λ-conversion and existential closure (Heim, 1982), we propose a semantics for (19) as in (21).

(21) $[\![(19)]\!] =$
$\exists e[\text{complain}_p(e) \wedge \text{agent}_p(e,\ \text{peter}) \wedge \exists e' \subseteq e\ \exists e'' \subseteq e[\text{content}_p(e',\ [\![\text{that he again had to pay for the whole group}]\!]) \wedge \text{SIM}_{p*}(\text{form}(e''),\ \text{pouting emoji})]]$

Thus, according to (21), the sentence in (19) is true, iff there exists a complaining event where Peter is the agent (= the person complaining) and there exists a subevent such that its content is the same as the proposition embedded under *complain* in (19) and there exists a further subevent—the facial expression Peter made during the speech event—such that its form is similar to the pouting emoji used in the ID report in (19) (cf. Maier's 2023 analysis of face emoji discussed above). In (21), only the emoji is what saturates the demonstration argument we posit for *complain*. In other words, the only quoted part in (19) is the emoji. The propositional variables indicate that only the last conjunct, i.e., the SIM predicate imposing the form constraint, is not-at-issue. All remaining parts make at-issue contributions, i.e., are on the table for discussion (Farkas and Bruce, 2010). This is reflected by the judgments in (15).

As briefly outlined above already, this proposal can be easily generalized to shifted interpretation of temporal and local adverbs and also to other iconic enrichments in ID receiving an interpretation from the perspective of the matrix subject beside face emoji: once such a shifted interpretation arises, the shifted elements are analyzed as quotes and thus saturate the demonstration, where their form resembles the words, phrases, or gestures, produced in the reported speech event.

An advantage of the analysis we propose is that it makes very clear predictions about the elements available for quotation in an ID utterance. Since the content of the subevent e' must be a proposition, only optional material can saturate the demonstration argument of a verb of saying like *complain*. Assuming now that all material used in an ID utterance can only be part either of e' or e'', the predictions are as follows. Only elements like temporal and local deictic adverbials, gestures, emoji, or the intonation used in the speech event that is reported are available for quotation in ID. Importantly, this also explains why first- and second-person pronouns cannot receive shifted interpretations in ID (at least not in German, English, and many other languages): since they are obligatory material which consequently needs to saturate the proposition argument of a verb of saying, they cannot at the same time saturate the demonstration argument and hence cannot be part of the quotation.

The analysis we propose, however, also raises certain questions. For example, from a syntactic point of view, when receiving a shifted interpretation, temporal and local deictic expressions would have to be moved covertly to a peripheral position in order to be able to saturate the demonstration argument of a verb of saying. Additionally, the account predicts that quoted constituents in ID always make not-at-issue contributions by default, since they enter a not-at-issue similarity relation with the form of a subevent of the reported speech event. While this is an interesting prediction, this still needs to be tested experimentally in future research, especially for the case of shifted adverbials.

5. Conclusion

In this paper, we investigated whether face emoji in ID can receive a shifted interpretation, meaning that they reflect the emotional stance of the matrix subject rather than the speaker. While previous research suggested that emoji are typically author-oriented, our experimental study provided evidence that their interpretation can shift when the perspective of the matrix subject is made prominent in the discourse. We analyzed this phenomenon within a mixed-quotational framework, arguing that certain elements in ID, such as emoji, function as unmarked quotations that demonstrate aspects of the reported speech event.

To support our claim, we conducted the rating study reported in Section 3, manipulating whether the matrix subject's or the speaker's perspective was emphasized in ID. Our results showed that sentences in which the matrix subject's perspective was more prominent received higher acceptability ratings when accompanied by a shifted emoji. This suggests that, under the right conditions, face emoji in ID can be interpreted as depicting the emotions of the matrix subject rather than the speaker.

Building on these findings, we proposed the formal analysis in Section 4, treating shifted face emoji as demonstrations in the sense of (Clark and Gerrig, 1990), where they iconically depict elements of the reported speech event. We extended this approach to other perspective-dependent expressions, such as temporal adverbials and gestures, arguing that their shiftability in ID can be understood within a broader account of mixed quotation, thereby challenging traditional analyses of ID arguing that it is strictly non-quotational.

A potential point of criticism for our formal treatment of ID is that we currently assume that demonstration and thus quotation is a constant feature of ID utterances. While shifted interpretations of perspective-dependent expressions occur more frequently than often assumed in ID—which we analyze as quotations—they still do not seem to be the norm. Thus, the account overgenerates to some extent. An appealing alternative approach could be to posit a more standard lexical entry for verbs of saying, i.e., one that is non-quotational in the default case. A possibility was sketched in (18c), where a verb of saying only takes a proposition, an individual, and an event as its arguments. This proposal strongly relies on insights from Maier (2017). If, then, a lexical element or an emoji, for example, receives a shifted interpretation and consequently quotation becomes involved, the quotational lexical entry as stated in (10) arises through a type shifting operation. Since this would be quite a complex operation, we argue that it can only arise in cases where the quotation is appropriately pragmatically marked. Spelling this idea out in more detail is left to future research.

Our proposal leaves open some further appealing avenues for future research. First, it remains unclear which perspective-dependent expressions can shift in ID and if so, at what ease they can do so. To our knowledge, a systematic investigation of this is lacking to date (but see, e.g., Anderson, 2019 for an experimental investigation of the shiftability of the temporal adverbial *tomorrow* in English). Moreover, it is unclear which factors beside perspectival prominence for face emoji and temporal adverbials promote shifted interpretations of perspective-dependent expressions in ID. More generally, this relates to the remark made above, since it is unclear when quotation in ID is sufficiently pragmatically marked. The prediction of our account that shifted interpretations of perspective-dependent expressions in ID make not-at-issue contributions should also be tested experimentally. Finally, our proposal should be extended to DD and FID in subsequent work, since this would allow for a comprehensive theory of speech reports, which is to our knowledge lacking in formal semantics. The common denominator of all speech report types, then, would be that they all at least allow for quotation, and some of them even require it.

More broadly, our proposal adds to existing research that different modes of speech reporting are often combined. Maier (2018), who draws a strict line between direct quotation and demonstration, notes that the two means of reporting often go hand in hand, i.e., more often than not they occur together. While we remain agnostic at this point whether a strict distinction between the two is desirable or not because research on demonstrations in formal semantics is scarce and their status thus largely undetermined, we still posit that in ID a combination of describing and depicting by means of a demonstration is available and probably also more persistent than often assumed in formal semantics. This insight adds to two things. First, it adds to the landscape of speech reporting proposed by Bary and Maier (2021) showing that different strategies of speech reporting can be combined. Second, and more generally, our proposal reflects a property of natural language often ignored in formal semantics: when turning to multimodal, iconic, or signed data, we can often observe that speakers and signers combine descriptive with depictive/iconic strategies, examples being iconic co-speech gestures (e.g., Ebert and Ebert, 2014; Schlenker, 2018; Esipova, 2019), ideophones (Barnes, 2024), classifiers in sign language (Schlenker and Lamberton, 2024), or, as in the case, of the present paper, face emoji.

References

Anand, P. and A. Nevins (2004). Shifty operators in changing contexts. In R. B. Young (Ed.), *Proceedings of Semantics and Linguistic Theory (SALT) 14*, Ithaca, NY, pp. 20–37. CLC Publications.

Anderson, C. J. (2019). Tomorrow isn't always a day away. In M. T. Espinal, E. Castroviejo, M. Leonetti, L. McNally, and C. Real-Puigdollers (Eds.), *Proceedings of Sinn und Bedeutung 23*, Barcelona, Spain, pp. 37–56. Universitat Autònoma de Barcelona.

Barnes, K. and C. Ebert (2023). The information status of iconic enrichments: Modelling gradient at-issueness. *Theoretical Linguistics 49*(3-4), 167–223.

Barnes, K., C. Ebert, R. Hörnig, and T. Stender (2022). The at-issue status of ideophones in German: An experimental approach. *Glossa: A Journal of General Linguistics 7*(1), 1–39.

Barnes, K. R. (2024). *The semantics and pragmatics of German ideophones: Warum geht der Frosch plitsch platsch die Treppe hoch?* Ph. D. thesis, Goethe University Frankfurt, Frankfurt, Germany.

Bary, C. and E. Maier (2021). The landscape of speech reporting. *Semantics and Pragmatics 14*(8), 1–54.

Cappelen, H. and E. Lepore (1997). Varieties of quotation. *Mind 106*(423), 429–450.

Christensen, R. H. B. (2023). *ordinal—Regression Models for Ordinal Data*. R package version 2023.12-4.

Clark, H. H. and R. J. Gerrig (1990). Quotations as demonstrations. *Language 66*(4), 764–805.

Davidson, D. (1979). Quotation. *Theory and Decision 11*(1), 27.

Davidson, K. (2015). Quotation, demonstration, and iconicity. *Linguistics and Philosophy 38*(6), 477–520.

De Brabanter, P. (2017). Why quotation is not a semantic phenomenon, and why it calls for a pragmatic theory. In I. Depraetere and R. Salie (Eds.), *Semantics and Pragmatics: Drawing a Line*, pp. 227–254. Cham, Switzerland: Springer.

De Brabanter, P. (2023). Quotation does not need marks of quotation. *Linguistics 61*(2), 285–316.

Ebert, C. and C. Ebert (2014). Gestures, demonstratives, and the attributive/referential distinction. Talk given at *Semantics and Philosophy in Europe 7*.

Ebert, C., C. Ebert, and R. Hörnig (2020). Demonstratives as dimension shifters. In *Proceedings of Sinn und Bedeutung 24*, Osnabrück, pp. 161–178. University of Osnabrück.

Ebert, C. and S. Hinterwimmer (2022). Free indirect discourse meets character viewpoint gestures. In S. Featherston, R. Hörnig, A. Konietzko, and S. von Wietersheim (Eds.), *Proceedings of Linguistic Evidence 2020: Linguistic Theory Enriched by Experimental Data*, Tübingen, Germany, pp. 333–349. University of Tübingen.

Esipova, M. (2019). *Composition and Projection in Speech and Gesture*. Ph. D. thesis, New York University, New York, NY.

Farkas, D. F. and K. B. Bruce (2010). On reacting to assertions and polar questions. *Journal of Semantics 27*(1), 81–118.

Geurts, B. (1999). *Presupposiitons and Pronouns*. Oxford, UK: Elsevier.

Greenberg, G. (2013). Beyond resemblance. *Philosophical Review 122*(2), 215–287.

Greenberg, G. (2021). Semantics of pictorial space. *Review of Philosophy and Psychology 12*(4), 847–887.

Grosz, P. G. (to appear). Expressivity and emojis. In D. Gutzmann and K. Turgay (Eds.), *The*

Oxford Handbook of Expressivity.

Grosz, P. G., G. Greenberg, C. De Leon, and E. Kaiser (2023). A semantics of face emoji in discourse. *Linguistics and Philosophy 46*(4), 905–957.

Grosz, P. G., E. Kaiser, and F. Pierini (2021). Discourse anaphoricity and first-person indexicality in emoji resolution. In P. G. Grosz, L. Martí, H. Pearson, Y. Sudo, and S. Zobel (Eds.), *Proceedings of Sinn und Bedeutung 25*, pp. 340–357.

Hacquard, V. (2010). On the event relativity of modal auxiliaries. *Natural Language Semantics 18*(1), 79–114.

Heim, I. (1982). *The Semantics of Definite and Indefinite Noun Phrases*. Ph. D. thesis, University of Massachusetts Amherst, Amherst, MA.

Heim, I. (1983). On the projection problem for presuppositions. In M. Barlow, D. P. Flickinger, and M. T. Wescoat (Eds.), *Proceedings of the Second West Coast Conference on Formal Linguistics*, Stanford, CA, pp. 114–125. Stanford Linguistics Association.

Henderson, R. (2016). A demonstration-based account of (pluractional) ideophones. In M. Moroney, C.-R. Little, J. Collard, and D. Burgdorf (Eds.), *Proceedings of Semantics and Linguistic Theory (SALT) 26*, Austin, TX, pp. 664–683.

Hintikka, K. J. J. (1969). Semantics for propositional attitudes. In J. W. Davis, D. J. Hockney, and W. K. Wilson (Eds.), *Philosophical Logic*, pp. 21–45. Dordrecht, The Netherlands: Reidel.

Johnson, M. (2017). Quotation through history: A historical case for the proper treatment of quotation. In P. Saka and M. Johnson (Eds.), *The Semantics and Pragmatics of Quotation*, pp. 281–302. Dordrecht, The Netherlands: Springer.

Kaiser, E. and P. G. Grosz (2021). Anaphoricity in emoji: An experimental investigation of face and non-face emoji. In P. Farrell (Ed.), *Proceedings of the Linguistic Society of America 6*, pp. 1009–1023.

Kirk-Giannini, C. D. (2024). Covert mixed quotation. *Semantics and Pragmatics 17*(5), 1–54.

Kratzer, A. (2006). Decomposing attitude verbs. Talk given at the workshop *Honoring Anita Mittwoch on her 80th birthday*.

Kratzer, A. (2016). Evidential moods in attitude and speech reports. Talk given at *UConn*.

Langendoen, D. T. and H. Savin (1971). The projection problem for presuppositions. In C. J. Fillmore and D. T. Langendoen (Eds.), *Studies in Linguistic Semantics*, pp. 54–60. New York, NY: Irvington.

Maier, E. (2015). Quotation and unquotation in free indirect discourse. *Mind & Language 30*(3), 345–373.

Maier, E. (2017). The pragmatics of attraction. In P. Saka and M. Johnson (Eds.), *The Semantics and Pragmatics of Quotation*, pp. 259–280. Dordrecht, The Netherlands: Springer.

Maier, E. (2018). Quotation, demonstration, and attraction in sign language role shift. *Theoretical Linguistics 44*(3-4), 265–276.

Maier, E. (2020). Quotes as complements: A Kratzerian approach. In R. Bhatt, I. Frana, and P. Menéndez-Benito (Eds.), *Making Worlds Accessible: Essays in Honor of Angelika Kratzer*, pp. 91–101. Amherst, MA: UMass Amherst.

Maier, E. (2023). Emojis as pictures. *Ergo: An Open Access Journal of Philosophy 10*(11), 310–340.

Pierini, F. (2021). Emojis and gestures: A new typology. In P. G. Grosz, L. Martí, H. Pearson, Y. Sudo, and S. Zobel (Eds.), *Proceedings of Sinn Und Bedeutung 25*, London, UK, pp.

720–732.
Plank, F. (1986). Über den Personenwechsel und den anderer deiktischer Kategorien in indirekter Rede. *Zeitschrift für germanistische Linguistik 14*(3), 284–308.
Potts, C. (2005). *The Logic of Conventional Implicatures*. Oxford, UK: Oxford University Press.
Potts, C. (2007). The dimensions of quotation. In C. Barker and P. Jacobson (Eds.), *Direct Compositionality*, pp. 405–431. Oxford, UK: Oxford University Press.
Quine, W. V. (1940). *Mathematical Logic*, Volume 4. Cambridge, MA: Harvard University Press.
R Core Team (2022). *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing.
Recanati, F. (2001). Open quotation. *Mind 110*(439), 637–687.
Schlenker, P. (2003). A plea for monsters. *Theoretical Linguistics 26*(1), 29–120.
Schlenker, P. (2004). Context of thought and context of utterance: A note on free indirect discourse and the historical present. *Mind & Language 19*(3), 279–304.
Schlenker, P. (2018). Gesture projection and cosuppositions. *Linguistics and Philosophy 41*(3), 295–365.
Schlenker, P. and J. Lamberton (2024). Iconological semantics. *Linguistics and Philosophy 47*, 779–838.
Shanon, B. (1976). On the two kinds of presuppositions in natural language. *Foundations of Language 14*(2), 247–249.
Sharvit, Y. (2008). The puzzle of free indirect discourse. *Linguistics and Philosophy 31*(3), 353–395.
Simons, M., J. Tonhauser, D. Beaver, and C. Roberts (2010). What projects and why. In N. Li and D. Lutz (Eds.), *Proceedings of Semantics and Linguistic Theory (SALT) 20*, Ithaca, NY, pp. 309–327. CLC Publications.
Stec, K. (2012). Meaningful shifts: A review of viewpoint markers in co-speech gesture and sign language. *Gesture 12*(3), 327–360.
Stec, K. (2016). *Visible quotation: The multimodal expression of viewpoint*. Ph. D. thesis, Rijksuniversiteit Groningen, Groningen, The Netherlands.
Steinbach, M. (2023). Angry lions and scared neighbors: Complex demonstrations in sign language role shift at the sign-gesture interface. *Linguistics 61*(2), 391–416.
Stokke, A. (2021). Protagonist projection, character focus and mixed quotation. In E. Maier and A. Stokke (Eds.), *The Language of Fiction*, pp. 379–404. Oxford, UK: Oxford University Press.
Syrett, K. and T. Koev (2015). Experimental evidence for the truth conditional contribution and shifting information status of appositives. *Journal of Semantics 32*(3), 525–577.
Tonhauser, J. (2012). Diagnosing (not-)at-issue content. In E. Bogal-Allbritten (Ed.), *Proceedings of Semantics of Under-Represented Languages of the Americas (SULA) 6*, pp. 239–254.
Umbach, C. and H. Gust (2014). Similarity demonstratives. *Lingua 149*, 74–93.
von Fintel, K. (2004). Would you believe it? The king of France is back! In A. Bezuidenhout and M. Reimer (Eds.), *Descriptions and Beyond*, pp. 315–347. Oxford, UK: Oxford University Press.
Wade, E. and H. H. Clark (1993). Reproduction and demonstration in quotations. *Journal of Memory and Language 32*(6), 805–819.

Walter, S. (2025a). Indirect discourse as mixed quotation? An experimental investigation. In T. Knowlton, F. Schwarz, and A. Papafragou (Eds.), *Proceedings of Experiments in Linguistic Meaning (ELM) 3*, Philadelphia, PA, pp. 423–434.

Walter, S. (2025b). A mixed-quotational account of indirect discourse: Evidence from self-pointing gestures. In Y. Zhang, F. L. Zhao, Y. Cho, and Y. Wu (Eds.), *Proceedings of Semantics and Linguistic Theory (SALT) 34*, Rochester, NY, pp. 88–109.

Wickham, H., M. Averick, J. Bryan, W. Chang, L. D. McGowan, R. François, G. Grolemund, A. Hayes, L. Henry, J. Hester, M. Kuhn, T. L. Pedersen, E. Miller, S. M. Bache, K. Müller, J. Ooms, D. Robinson, D. P. Seidel, V. Spinu, K. Takahashi, D. Vaughan, C. Wilke, K. Woo, and H. Yutani (2019). Welcome to the tidyverse. *Journal of Open Source Software 4*(43), 1686.